ASSIGNMENT 4.1

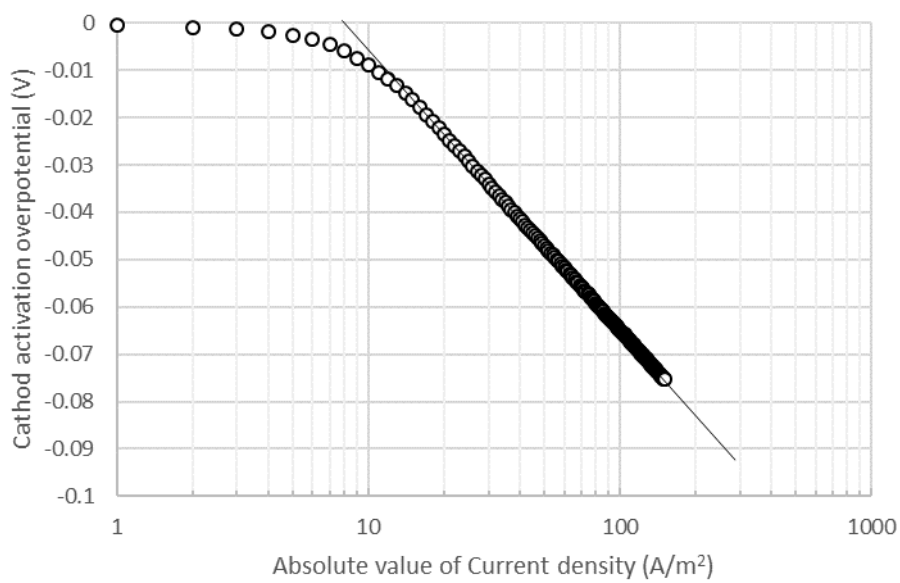
Q1 Tafel Slope Consider the polarization data for the cathode of a fuel cell given in the attached spreadsheet. The experiment is run at 25ºC (298.15 K) under atmospheric pressure (1 atm). Using the data, determine:

a) The reference exchange current density, I_o and
b) Cathodic transfer coefficient, α_c

Note: For a cathode, net current is in backwards(cathodic) direction therefore $I < 0$ and the Tafel expression is:

$$\eta = A - B\log(-I) \;\; where\; A = 2.303\frac{\overline{R}T}{\alpha_c F}log(I_0) \;\; and\; B = 2.303\frac{\overline{R}T}{\alpha_c F}$$

You should fit a linear line only to the linear part of Tafel curve. Recall that Tafel curve is not applicable for small current densities (e.g. $|I| \gg I_o$)



Q2 Ohmic Overpotential Determine potential drop (ohmic polarization) across a 100 µm thick electrolyte with a conductivity of 10 S/m if the current density is 1 A/cm^2.

Q3 Area Specific Resistance In early 2000s, polymer electrolyte fuel cells used Nafion® 117, which is 7 mil (0.007 in) thick. By early 2010s, the improvement in durability and manufacturing processes made 20 um (0.020 mm) thick membranes, e.g. Nafion® HP feasible. Assuming the ionic conductivity did not change (assume constant at 0.05 S/cm), determine the % decrease in the area specific resistance. What would be the decrease in ohmic overpotential for a fuel cell operated at 1 A/cm^2 if it used a Nafion® HP instead of Nafion® 117.

Q4 Faraday's Law Calculate the hydrogen flow rate (in mol/s) required for 1000 A of external current if all non-idealities (including hydrogen cross-over and electronic short) can be ignored.
Faraday's law: $\dot{n} = i/(nF)$